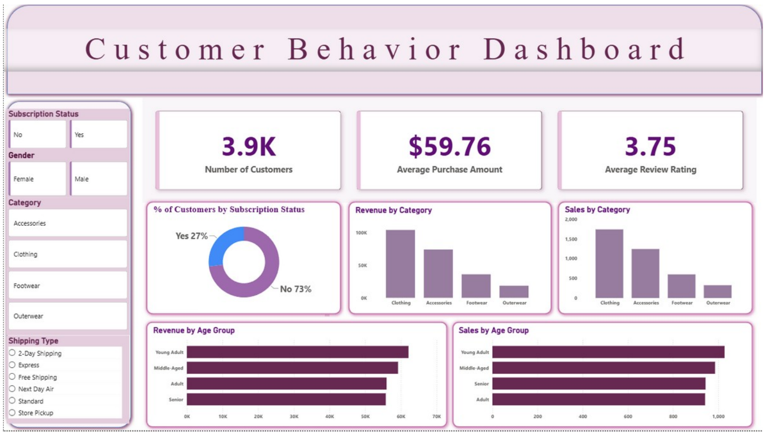
CHANDIGARH UNIVERSITY

Department of Computer Science & Engineering

A PROJECT REPORT ON

Customer Shopping Behavior Analysis:

A Customer Behavior Dashboard using Python, MySQL & Power BI



Submitted By

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B.Tech, Computer Science & Engineering

Academic Year: 2026

Certificate

This is to certify that the project report entitled “Customer Shopping Behavior Analysis: A Customer Behavior Dashboard using Python, MySQL & Power BI” is an independent work carried out by Devansh Singh Raghuvanshi, in partial fulfilment of the requirements for the completion of a self-directed project in the field of Data Analytics and Business Intelligence, during the academic year 2026.

The work presented in this report is the result of the candidate's own effort in dataset analysis, data engineering, database integration, and dashboard development, and has not been submitted elsewhere for the award of any other degree or diploma.

Project Guide / Supervisor:

____

Department of Computer Science & Engineering
Chandigarh University

Declaration

I, Devansh Singh Raghuvanshi, hereby declare that the project report titled “Customer Shopping Behavior Analysis: A Customer Behavior Dashboard using Python, MySQL & Power BI” is an original and authentic record of the work carried out by me.

All data, source code, analysis, and dashboard visuals presented in this report were developed independently, using publicly available tools, libraries, and documentation referenced in the Bibliography section. Any content adapted from external sources has been duly acknowledged.

This report has not been submitted, in part or in full, to any other institution for the award of any degree, diploma, or certificate.

Devansh Singh Raghuvanshi

B.Tech, Computer Science & Engineering
Chandigarh University — Academic Year 2026

Acknowledgement

I would like to express my sincere gratitude to the faculty of the Department of Computer Science & Engineering, Chandigarh University, for their guidance, encouragement, and valuable feedback throughout the course of this project.

I am thankful for the open-source community behind Python, Pandas, SQLAlchemy, MySQL, and Microsoft Power BI, whose extensive documentation made it possible to design and implement a complete end-to-end analytics pipeline.

Finally, I extend my appreciation to my family and peers for their continuous support and motivation during the development of this project.

Abstract

Retail businesses generate large volumes of customer transaction data, yet many organizations struggle to convert this raw data into actionable insight. This project presents an end-to-end Business Intelligence solution — the Customer Behavior Dashboard — that transforms a raw customer shopping dataset of approximately 3,900 records into a fully interactive analytics dashboard.

The solution follows a complete ETL (Extract, Transform, Load) methodology implemented in Python using the Pandas library. Missing values in customer review ratings were imputed using category-wise median values, column names were standardized, and new engineered features — including customer age groups and numerical purchase-frequency-in-days — were derived to enrich the dataset. The cleaned dataset was then loaded into a MySQL relational database using SQLAlchemy and PyMySQL, providing a persistent and queryable data layer.

A Power BI dashboard was subsequently developed on top of the MySQL data source, featuring KPI cards, donut charts, bar charts, and interactive slicers for subscription status, gender, category, and shipping type. The resulting dashboard reveals that Clothing is the highest-performing product category by both revenue and sales volume, that Young Adult and Middle-Aged customers are the strongest-contributing demographic segments, and that only 27% of customers are currently subscribed — highlighting a significant opportunity for loyalty-based marketing.

This report documents the complete technical workflow, system architecture, data design, and business findings of the project, and concludes with actionable recommendations for retail decision-makers.

Keywords: *Business Intelligence, ETL, Python, Pandas, MySQL, SQLAlchemy, Power BI, Customer Analytics, Data Visualization*

Table of Contents

Chapter 1: Introduction

1.1 Background

In the modern retail landscape, understanding customer purchasing behavior is critical to designing effective marketing strategies, optimizing inventory, and improving customer retention. Every customer interaction — a purchase, a review, a subscription choice — generates data that, when properly structured and analyzed, can reveal patterns that are invisible in raw transactional form.

This project analyzes a retail customer shopping dataset to uncover behavioral patterns across product categories, age demographics, and subscription status, and presents these findings through an interactive Power BI dashboard supported by a Python-based data engineering pipeline and a MySQL database backend.

1.2 Problem Statement

Raw retail data is typically unstructured for direct decision-making: it contains missing values, inconsistent naming conventions, and features that are not immediately business-relevant (such as text-based purchase frequency). Without a structured ETL process and a visualization layer, business stakeholders cannot easily extract insight from this data.

1.3 Scope of the Project

- Clean and pre-process a real-world customer shopping dataset (~3,900 records).
- Engineer new analytical features such as age groups and numerical purchase frequency.
- Persist the cleaned dataset into a relational MySQL database.
- Design and build an interactive Power BI dashboard for business reporting.
- Derive actionable business recommendations from the resulting insights.

1.4 Project Objectives

- Clean and prepare the raw customer dataset.
- Handle missing values using statistically sound imputation methods.
- Create new useful features such as age group and purchase frequency in days.
- Store the cleaned data into a MySQL database using SQLAlchemy.
- Build an interactive customer behavior dashboard in Power BI.
- Identify key insights related to customers, sales, revenue, and product categories.
- Provide data-driven business recommendations.

Chapter 2: Tools and Technologies

The project uses a modern, industry-standard data analytics stack spanning data processing, database management, and visualization.

Tool / Technology	Purpose
Python 3.x	Core programming language for data analysis and processing
Pandas	Data cleaning, transformation, and feature engineering
SQLAlchemy	ORM layer connecting Python with MySQL
PyMySQL	MySQL database driver
MySQL Community Server 8.x	Persistent relational data storage
Power BI Desktop	Interactive dashboard design and visualization
VS Code	Development environment for Python scripting

Table 2.1: Tools and Technologies Used

Chapter 3: Dataset Overview

3.1 Dataset Description

The dataset used in this project, customer_shopping_behavior.csv, contains retail transaction and customer profile information across roughly 3,900 records and 15+ attributes of mixed numerical and categorical type.

Column Type	Description
Customer Details	Age, Gender
Purchase Details	Purchase Amount, Category, Frequency of Purchases
Customer Feedback	Review Rating
Marketing Details	Discount Applied, Promo Code Used, Subscription Status
Delivery Details	Shipping Type

Table 3.1: Dataset Column Groups

3.2 Loading and Initial Exploration

The dataset was loaded using Pandas, followed by an initial structural and statistical exploration:

```
df = pd.read_csv('customer_shopping_behavior.csv')

print(df.head())
print(df.info())
print(df.describe(include='all'))
print(df.isnull().sum())
```

These commands helped establish the structure, data types, missing-value counts, and summary statistics of the dataset prior to cleaning.

3.3 Numerical and Categorical Attributes

Attribute Type	Columns
Numerical	Age, Purchase Amount, Review Rating, Previous Purchases
Categorical	Gender, Category, Shipping Type, Subscription Status, Payment Method, Season

Table 3.2: Numerical vs. Categorical Attributes

Chapter 4: System Design and Architecture

This chapter presents the architectural and modeling diagrams that describe how data flows through the system, from raw CSV ingestion to the final Power BI dashboard.

4.1 System Architecture

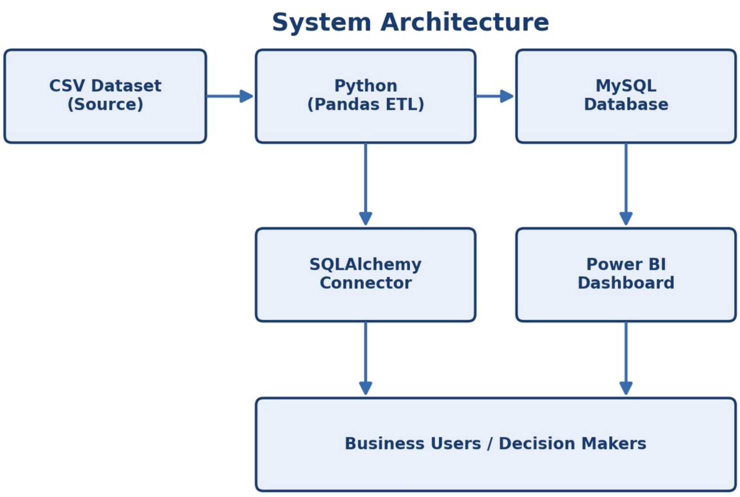

Figure 4.1: System Architecture

4.2 Project Workflow

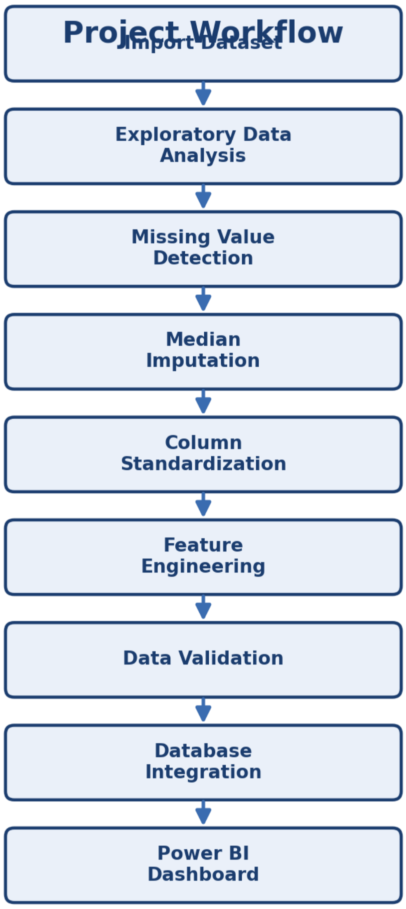

Figure 4.2: End-to-End Project Workflow

4.3 Data Flow Diagram – Level 0

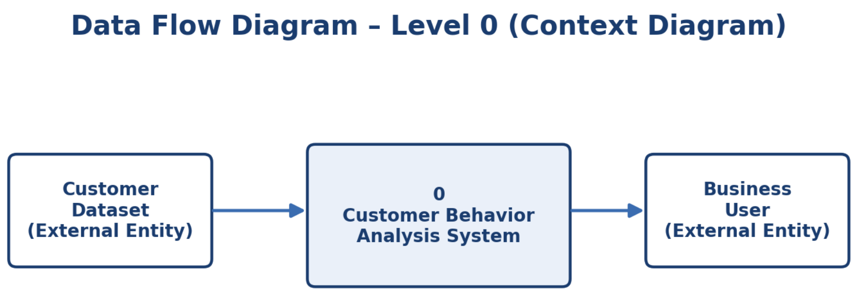

Figure 4.3: DFD Level 0 (Context Diagram)

4.4 Data Flow Diagram – Level 1

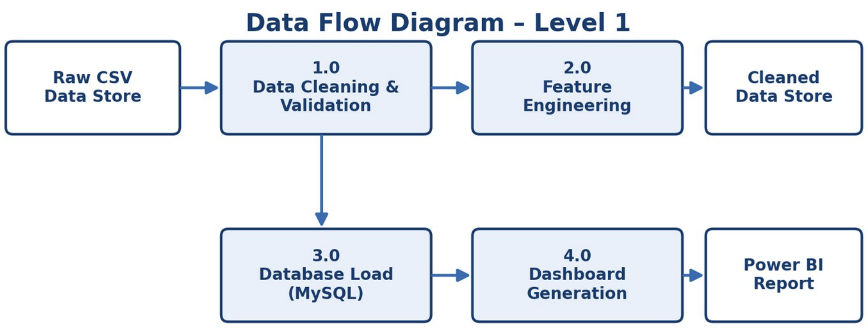

Figure 4.4: DFD Level 1

4.5 Use Case Diagram

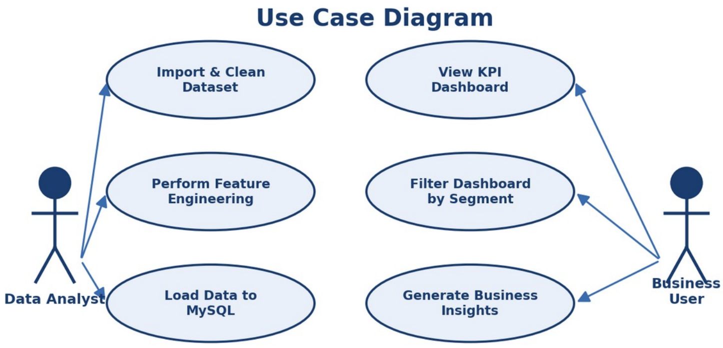

Figure 4.5: Use Case Diagram

4.6 Activity Diagram

Figure 4.6: Activity Diagram

4.7 Entity Relationship (ER) Diagram

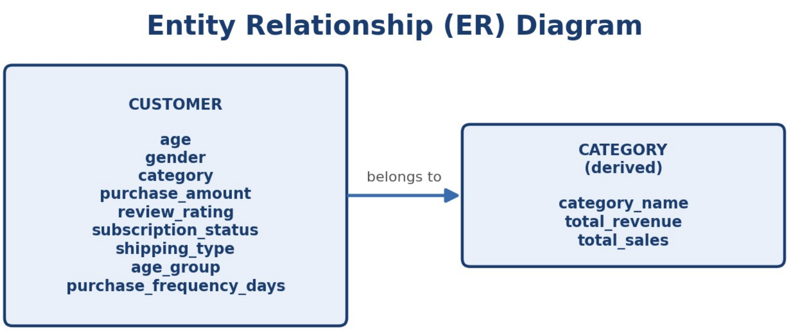

Figure 4.7: ER Diagram for the Customer Table

Chapter 5: Data Cleaning and Transformation

5.1 Missing Value Handling

Missing values in the Review Rating column were filled using the median review rating within each product category, rather than a single global median, since rating patterns vary meaningfully across categories.

```
df['Review Rating'] = df.groupby('Category')['Review Rating'].transform(
    lambda x: x.fillna(x.median())
)
```

5.2 Column Name Standardization

Column names were standardized to lowercase, snake_case form to simplify downstream Python and SQL usage.

```
df.columns = df.columns.str.lower()
df.columns = df.columns.str.replace(' ', '_')
df = df.rename(columns={'purchase_amount_(usd)': 'purchase_amount'})
```

5.3 Redundant Column Removal

The discount_applied and promo_code_used columns were compared to check for redundancy:

```
print(df[['discount_applied', 'promo_code_used']].head(10))

isTrue = (df['discount_applied'] == df['promo_code_used']).all()
print(isTrue)

df = df.drop('promo_code_used', axis=1)
```

Since the two columns carried overlapping information, promo_code_used was dropped from the final dataset to avoid redundancy.

Chapter 6: Feature Engineering

6.1 Age Group Segmentation

Customers were segmented into four quartile-based age groups to support demographic analysis.

```
labels = ['Young Adult', 'Adult', 'Middle-Aged', 'Senior']
df['age_group'] = pd.qcut(df['age'], q=4, labels=labels)
```

Age Group
Young Adult
Adult
Middle-Aged
Senior

Table 6.1: Age Group Categories

6.2 Purchase Frequency Conversion

The text-based frequency_of_purchases column was converted into a numerical purchase_frequency_days column to enable quantitative frequency analysis.

```
frequency_mapping = {
    'Fortnightly': 14, 'Weekly': 7, 'Monthly': 30,
    'Quaterly': 90, 'Bi-Weekly': 14, 'Annually': 365,
    'Every 3 Months': 90
}
df['purchase_frequency_days'] = df['frequency_of_purchases'].map(frequency_mapping)
```

Frequency of Purchases	Days
Weekly	7
Fortnightly	14
Bi-Weekly	14
Monthly	30
Quarterly / Every 3 Months	90
Annually	365

Table 6.2: Purchase Frequency Mapping

Chapter 7: Database Integration

7.1 ETL Pipeline

Figure 7.1: Extract-Transform-Load (ETL) Pipeline

7.2 Database Connection

A MySQL connection was established using SQLAlchemy and the PyMySQL driver.

```
from sqlalchemy import create_engine
from urllib.parse import quote_plus

username = "root"
password = quote_plus("********")
host = "127.0.0.1"
port = "3306"
database = "mydatabase"

engine = create_engine(
    f"mysql+pymysql://{username}:{password}@{host}:{port}/{database}"
)
```

7.3 Writing and Verifying Data

```
table_name = "customer"
df.to_sql(table_name, engine, if_exists="replace", index=False)
print("Data inserted successfully!")

result = pd.read_sql("SELECT * FROM customer LIMIT 10;", engine)
print(result)
```

The if_exists="replace" option ensures the table is refreshed with the latest cleaned dataset on every run, and the sample read-back confirmed successful insertion.

Chapter 8: Dashboard Design and Implementation

8.1 Dashboard Architecture

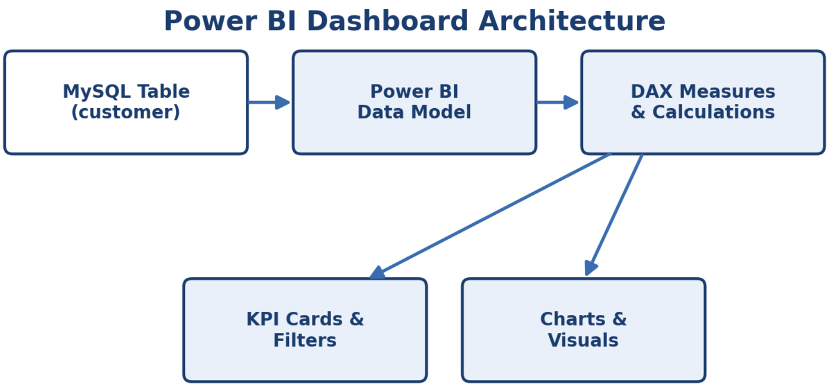

Figure 8.1: Power BI Dashboard Architecture

8.2 Dashboard KPIs

KPI	Value
Number of Customers	3.9K
Average Purchase Amount	$59.76
Average Review Rating	3.75

Table 8.1: Dashboard Key Performance Indicators

8.3 Dashboard Filters

- Subscription Status
- Gender
- Category
- Shipping Type

8.4 Dashboard Screenshot

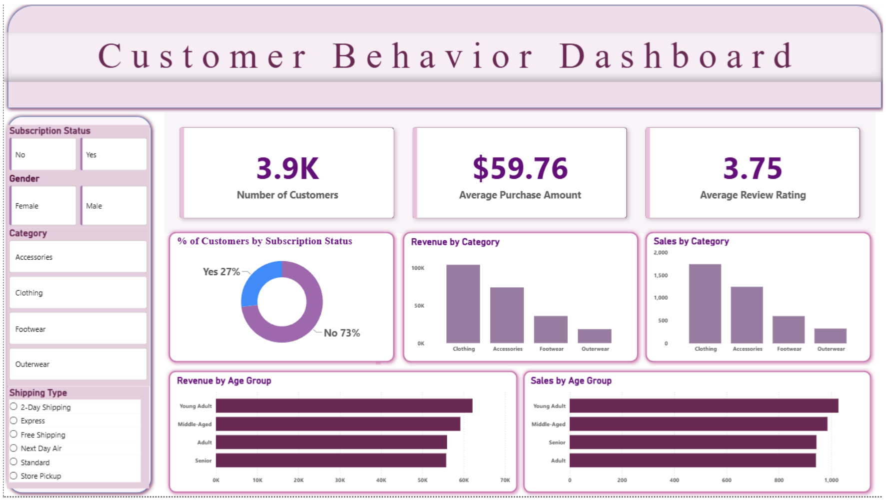

Figure 8.2: Customer Behavior Dashboard (Power BI)

Chapter 9: Key Insights

9.1 Subscription Status

Subscription Status	Percentage
No	73%
Yes	27%

Table 9.1: Subscription Status Breakdown

Insight: Most customers are not subscribed. Only 27% of customers hold a subscription, indicating a strong opportunity to increase subscription adoption through targeted offers and loyalty programs.

9.2 Revenue and Sales by Category

Revenue and sales are led by Clothing, followed by Accessories, Footwear, and Outerwear, in that order for both metrics.

Insight: Clothing is the strongest category by both sales volume and revenue, making it the primary category for business focus.

9.3 Revenue and Sales by Age Group

Young Adult and Middle-Aged customers generate the highest revenue and sales, while Adult and Senior customers show comparatively lower activity.

Insight: Younger customer segments are the most active shoppers and the most valuable demographic for targeted campaigns.

Chapter 10: Business Recommendations and Conclusion

10.1 Improve Subscription Conversion

- Offer subscription discounts.
- Provide exclusive member-only deals.
- Introduce loyalty reward points.
- Promote free shipping for subscribers.
- Send personalized subscription offers to frequent buyers.

10.2 Focus More on the Clothing Category

- Maintain sufficient stock for clothing products.
- Run promotional campaigns for clothing.
- Introduce seasonal clothing offers.
- Cross-sell accessories with clothing purchases.

10.3 Target Young Adult and Middle-Aged Customers

- Create targeted marketing campaigns for these age groups.
- Use social media promotions.
- Provide personalized product recommendations.
- Offer bundle discounts based on shopping history.

10.4 Increase Engagement for Adults and Seniors

- Provide easy-to-use offers and product recommendations.
- Promote comfort-based or practical products.
- Offer simple discount campaigns.
- Improve communication through email or SMS marketing.

10.5 Use Purchase Frequency for Customer Retention

- Send reminders to customers who have not purchased recently.
- Provide special discounts to repeat customers.
- Identify high-frequency customers for loyalty programs.
- Create win-back campaigns for inactive customers.

10.6 Conclusion

This project successfully completed an end-to-end customer shopping behavior analysis, from raw data ingestion through to a fully interactive Power BI dashboard. The pipeline handled missing review ratings, standardized column names, engineered new demographic and frequency features, and persisted the cleaned dataset into a MySQL database using SQLAlchemy and PyMySQL.

The resulting dashboard shows that Clothing is the best-performing product category, that Young Adult and Middle-Aged customers are the strongest customer segments, and that subscription adoption remains relatively low at 27%. These insights, together with the accompanying business recommendations, provide a practical foundation for improving marketing strategy, customer retention, and data-driven decision-making within a retail organization.

The modular architecture developed in this project — a clean separation between data engineering (Python/Pandas), storage (MySQL/SQLAlchemy), and visualization (Power BI) — also provides a strong foundation for future enhancements such as predictive analytics, customer churn modeling, and cloud-based deployment.

References / Bibliography

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Official Documentation

- Python Documentation — https://docs.python.org/
- Pandas Documentation — https://pandas.pydata.org/docs/
- SQLAlchemy Documentation — https://docs.sqlalchemy.org/
- MySQL Documentation — https://dev.mysql.com/doc/
- Power BI Documentation — https://learn.microsoft.com/power-bi/
- PyMySQL Documentation — https://pymysql.readthedocs.io/

Online Learning Resources

- Microsoft Learn — Power BI Learning Resources
- Kaggle Dataset Repository
- GitHub Documentation
- Stack Overflow
- W3Schools SQL Tutorials
- Real Python
- Analytics Vidhya
- Towards Data Science

Appendices

Appendix A — Software Used

Software	Version
Python	3.x
Pandas	Latest Stable
SQLAlchemy	Latest Stable
PyMySQL	Latest Stable
MySQL Community Server	8.x
Power BI Desktop	Latest
VS Code	Latest

Appendix B — Hardware Configuration

Component	Specification
Processor	Intel Core i5 or Higher
RAM	8 GB
Operating System	Windows 10 / Windows 11
Storage	20 GB Available Space
Internet	Required for installation

Appendix C — Dataset Description

Attribute	Detail
Dataset Name	Customer Shopping Behavior Dataset
Dataset Size	Approximately 3,900 Records
Total Attributes	15+
Data Type	Mixed (Numerical & Categorical)

Appendix D — Python Workflow

- Import Dataset → Exploratory Data Analysis → Missing Value Detection
- Median Imputation → Column Standardization → Feature Engineering
- Data Validation → Database Integration → Power BI Dashboard

Appendix E — SQL Database

Item	Detail
Database Name	mydatabase

Item	Detail
Table Name	customer
Database Engine	MySQL
Python Connector	SQLAlchemy

Appendix F — Power BI Dashboard Components

- KPI Cards
- Donut Chart
- Bar Charts
- Interactive Filters
- Customer Demographics
- Category Analysis
- Revenue Analysis
- Sales Analysis

Appendix G — Business KPIs

KPI	Description
Total Customers	Number of Customers
Average Purchase Amount	Customer Spending
Average Review Rating	Customer Satisfaction
Revenue by Category	Category Performance
Sales by Category	Product Performance
Subscription Status	Customer Loyalty
Revenue by Age Group	Demographic Revenue
Sales by Age Group	Customer Segmentation

Appendix H — Learning Outcomes

Area	Skill Developed
Programming	Python
Data Analysis	Pandas
Database	MySQL
ORM	SQLAlchemy
Business Intelligence	Power BI
Data Visualization	KPI Dashboard
Feature Engineering	Customer Segmentation
Dashboard Design	Interactive Reporting

Appendix I — Project Achievements

- Data Exploration
- Data Cleaning
- Missing Value Handling
- Feature Engineering
- Data Validation
- SQL Database Integration
- Dashboard Development
- Business Intelligence
- Customer Analytics
- Decision Support System

Project Highlights

Attribute	Detail
Project Domain	Business Intelligence
Industry	Retail Analytics
Programming Language	Python
Database	MySQL
Visualization	Power BI
Dataset Size	3,900 Customers
Dashboard Type	Interactive
Analytics Type	Descriptive Analytics
Methodology	ETL Pipeline
Database Connectivity	SQLAlchemy